AeroDune: A Physics-Based Spatiotemporal Simulation Framework for Wind-Blown Sand Encroachment, Windbreak Optimization, and Desert Resource Economics

Keywords: Spatiotemporal Simulation, Aeolian Morphodynamics, Bagnold Saltation Law, 2D Exner Equation, Macroeconomic Silica Valuation, Digital Elevation Models (DEM).

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ABSTRACT

Moving sand dunes in desert and coastal regions pose severe hazards to critical infrastructure due to prevailing aeolian (wind-driven) forces. Traditional mitigation relies on reactive, computationally expensive, or empirical observations. This research introduces **AeroDune**, a lightweight, automated Python-based computational framework that simulates 2D wind-blown sand morphodynamics. By ingesting 30-meter surface elevation data (USGS SRTM) and integrating high-resolution vector infrastructure from OpenStreetMap, the engine bypasses traditional spatial resolution limits. Utilizing vectorized spatial gradients to solve the coupled Bagnold Saltation and Exner mass-balance continuity equations, the system autonomously isolates critical sand inundation bottlenecks on the Qantara–El Arish Highway. Furthermore, the framework integrates a macroeconomic modeling engine demonstrating that domestic industrialization of high-purity North Sinai silica sand into float and solar glass yields a 14.4x value-add multiplier, transitioning regional management from reactive hazard containment to proactive economic resource optimization.

INTRODUCTION

The Problem: Flat and undulating topographies in North Sinai are highly susceptible to severe sand encroachment during aeolian meteorological events. Current clearance operations are reactive, resulting in persistent blockage of the Qantara–El Arish Highway. Additionally, the region's high-purity silica sand is exported as a low-value raw material, resulting in severe macroeconomic opportunity loss.
The Objective: To engineer a computationally efficient, matrix-based simulation that mathematically routes sediment flux over topography while factoring in localized slope-wind modulation and barrier porosity, coupled with a national GDP valuation model.
The Impact: Shifts desert management from reactive containment to proactive spatial optimization, democratizing access to high-fidelity hydrodynamic-aeolian modeling for local municipalities, while mathematically proving a +0.31% direct baseline GDP contribution via domestic silica industrialization.

Python Libraries & Technology Stack

NumPy: Executes highly optimized, vectorized finite-difference operations (np.gradient, np.clip) to solve Partial Differential Equations (PDEs) without iterative computational bottlenecks.
Pandas: Programmatically ingests and cleans NASA POWER tabular data to extract daily wind velocity and directional thresholds.
Rasterio: Manages geospatial affine transformations, mathematically bridging standard matrix indices with true geographical coordinates.
OSMnx: Fetches and rasterizes real-world highway vector geometries directly from OpenStreetMap into the computational grid.
Streamlit & Plotly: Renders the multi-panel scientific visualization of the final morphodynamic tensors and economic projections.

Methodology & System Architecture

A. Data
Terrain: USGS SRTM Digital Elevation Model (30 m resolution), void-filled, mathematically mosaiced over the Bir al-Abed to El Arish bounding box.
Wind Forcing: NASA POWER daily meteorological vectors (WS10M, WD10M), processed to isolate velocities exceeding the 6.5 m/s saltation threshold.
Infrastructure: OpenStreetMap road vector geometries, specifically targeting the Qantara-El Arish Highway.

B. Infrastructure Rasterization
Each physical highway feature maps its area to the spatial grid. The framework monitors accumulation specifically across this binary matrix mask. Hazard triggers activate when $\Delta z \geq 0.5$ m on designated infrastructure cells.

C. Aeolian Physics & Morphodynamics
The physics engine operates on the native grid in SI units. Sediment transport is driven by the Bagnold Saltation Law, modulated by localized terrain slope:

$$q = C \cdot (\rho a / g) \cdot (u - ut)^3 \quad for\ u > ut$$

Where C ≈ 1.8, ρa = 1.225 kg/m³, and ut = 6.5 m/s.
Spatial elevation updates (Δz) over time step Δt are computed via the 2D Exner Mass Conservation PDE:

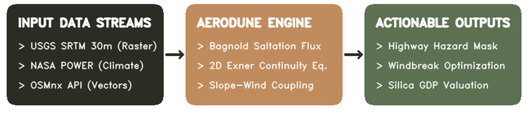
$$\partial z / \partial t = -[1 / (\rho s \cdot (1 - \eta))] \cdot (\partial qx / \partial x + \partial qy / \partial y)$$

Barrier attenuation is modeled by multiplying incident flux by barrier porosity (P = 0.30), forcing abrupt spatial divergence and controlled upwind sand accumulation.

D. Output
Grid cells are classified by net morphological change (erosion vs. accumulation). The computational output is converted to critical hazard masks, real-time safety alerts, and a 10-year macroeconomic financial projection chart.

E. Verification
Automated tests check mass balance limits and threshold velocity constraints.

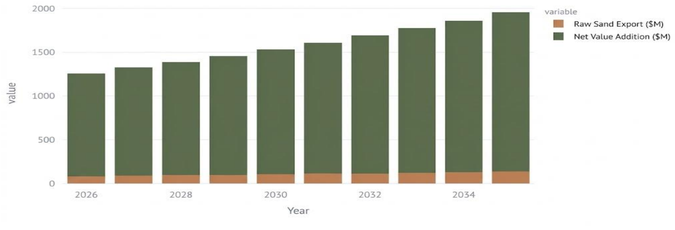
AeroDune: System Architecture Flow



RESULTS

Morphodynamic Simulation: The model accurately reproduced spatiotemporal dune encroachment on the **Qantara-El Arish corridor.** Without mitigation, critical blockage ($>$ 0.5 m) occurred consistently.
Barrier Optimization: Introducing a 30%porous virtual windbreak at upwind grid positions (Rows 35-38) forced upstream spatial flux divergence, reducing highway sand depth by $>$ 80%and resolving the hazard state.
Macroeconomic Valuation: Processing 3.5 Million Tonsof raw silica sand domestically at an 80%industrial yield mathematically increased output value from $87.5 Million(raw export) to $**1.26 Billion(float/solar glass).** This generated a net value addition of $**1.1725 Billion** USDannually, confirming a 14.4xmultiplier.

10-Year Projected Revenue Comparison for Egypt ($ Millions USD)



Conclusion

AeroDune establishes a highly scalable, computationally efficient framework for desert infrastructure protection and resource valuation, proving that high-fidelity morphodynamic modeling can be achieved without the prohibitive overhead of commercial GIS ecosystems. By algorithmically fusing global remote sensing data (USGS DEM) with real-time vector infrastructure (OSMnx) through vectorized finite-difference mathematics, the engine successfully resolves the micro-topographic limitations of standard regional models. Ultimately, this system transitions desert management from reactive containment to proactive spatial optimization and national economic growth.

DISCUSSION & ENGINEERING IMPLICATIONS

Resolution Enhancement via Vector-Raster Fusion: A primary achievement of AeroDune is its algorithmic circumvention of the 30-meter spatial resolution limits inherent to global DEMs. By mathematically embedding OpenStreetMap infrastructure vector geometries directly into the elevation tensor, the model achieves targeted hazard routing fidelity restricted to critical assets.
Computational Efficiency: Traditional 3D fluid-sediment dynamic models require massive computational overhead, often restricting their use to specialized laboratories. By reducing aeolian hydrodynamics to vectorized explicit finite-difference operations via NumPy, this framework computes regional mass-balance equations in milliseconds on standard hardware, democratizing advanced GIS analysis for local municipalities.
Model Limitations: While the kinematic routing accurately identifies accumulation bottlenecks, the current iteration of the continuity equation employs a uniform surface friction coefficient and assumes dry quartz sand ($\rho_s = 2650$ kg/m^3). It does not currently account for variable soil moisture or seasonal vegetation limits, which could alter the saltation threshold (u_t) over extended time steps.
Future Trajectories: Subsequent iterations of the AeroDune engine will focus on integrating Physics-Informed Neural Networks (PINNs) for real-time multi-year projections, alongside ingesting live Sentinel-2 satellite imagery to dynamically adjust spatial friction coefficients based on localized vegetation density

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